

Data for the Neurocomputing paper, including a
comparison with state-of-the-art methods and
images used in 3D objects.

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1 Appendix B. Comparison of state-of-the-art methods with our method using the 66 images from Sánchez-Cruz.

This document presents Tables 1, 2, and 3, which compare state-of-the-art methods with our approach for the 66 voxelized objects used in Experiment 3. The results demonstrate identical values for shape descriptors, simplexes, and tetravoxels. Only 12 objects from the sample are shown to verify the accuracy of these formulas, but the complete dataset can be requested via email at Cesareduardomucha@hotmail.com.

Table 1: Comparison between the state-of-the-art approach and the formulas discovered for each model. Part 1

Test images	Descriptors and Simplexes	Results	
		Traditional methods	All formulas found
Axolotl	Euler	-1	-1
	ES	248648	248648
	CS	9461726	9461726
	Volume	3195350	3195350
	Points	3321260	3321260
	Edges	9836285	9836285
	Faces	9710374	9710374
	Tetravoxels	0	0
Babydragon	Euler	-8	-8
	ES	61306	61306
	CS	768799	768799
	Volume	266484	266484
	Points	298021	298021
	Edges	861650	861650
	Faces	830105	830105
	Tetravoxels	739038	739038
Bird	Euler	-1	-1
	ES	21880	21880
	CS	301003	301003
	Volume	103981	103981
	Points	115438	115438
	Edges	334341	334341
	Faces	322883	322883
	Tetravoxels	290581	290581
Cheese	Euler	-2	-2
	ES	81148	81148
	CS	2143672	2143672
	Volume	728082	728082
	Points	769004	769004
	Edges	2265744	2265744
	Faces	2224820	2224820
	Tetravoxels	2103448	2103448

Table 2: Comparison between the state-of-the-art approach and the formulas discovered for each model. Part 2

Test images	Descriptors and Simplexes	Results	
		Traditional methods	All formulas found
Cup	Euler	0	0
	ES	5788	5788
	CS	12520	12520
	Volume	5138	5138
	Points	8120	8120
	Edges	21290	21290
	Faces	18308	18308
	Tetravoxels	9714	9714
Torus	Euler	0	0
	ES	6538	6538
	CS	26137	26137
	Volume	9802	9802
	Points	13294	13294
	Edges	36167	36167
	Faces	32675	32675
	Tetravoxels	23091	23091
Torush	Euler	-1	-1
	ES	6712	6712
	CS	25687	25687
	Volume	9681	9681
	Points	13255	13255
	Edges	35974	35974
	Faces	32399	32399
	Tetravoxels	22550	22550
Vase	Euler	0	0
	ES	20440	20440
	CS	34672	34672
	Volume	14964	14964
	Points	25226	25226
	Edges	65374	65374
	Faces	55112	55112
	Tetravoxels	24494	24494

Table 3: Comparison between the state-of-the-art approach and the formulas discovered for each model. Part 3

		Results	
Test images	Descriptors and Simplexes	Traditional methods	All formulas found
Bird	Euler	-1	-1
	ES	21880	21880
	CS	301003	301003
	Volume	103981	103981
	Points	115438	115438
	Edges	334341	334341
	Faces	322883	322883
	Tetravoxels	290581	290581
Elephant	Euler	1	1
	ES	40248	40248
	CS	646761	646761
	Volume	222295	222295
	Points	242931	242931
	Edges	707644	707644
	Faces	687009	687009
	Tetravoxels	627148	627148
Gun	Euler	-7	-7
	ES	23422	23422
	CS	215719	215719
	Volume	75810	75810
	Points	87922	87922
	Edges	251260	251260
	Faces	239141	239141
	Tetravoxels	204416	204416
Moon	Euler	1	1
	ES	77350	77350
	CS	3199273	3199273
	Volume	1079316	1079316
	Points	1118377	1118377
	Edges	3315683	3315683
	Faces	3276623	3276623
	Tetravoxels	3160983	3160983